



FEDERAL PUBLIC SERVICE COMMISSION
COMPETITIVE EXAMINATION-2019
FOR RECRUITMENT TO POSTS IN BS-17
UNDER THE FEDERAL GOVERNMENT

Roll Number

BOTANY

TIME ALLOWED: THREE HOURS	PART-I (MCQS)	MAXIMUM MARKS = 20
PART-I(MCQS): MAXIMUM 30 MINUTES	PART-II	MAXIMUM MARKS = 80
NOTE: (i) Part-II is to be attempted on the separate Answer Book. (ii) Attempt ONLY FOUR questions from PART-II. ALL questions carry EQUAL marks. (iii) All the parts (if any) of each Question must be attempted at one place instead of at different places. (iv) Write Q. No. in the Answer Book in accordance with Q. No. in the Q.Paper. (v) No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed. (vi) Extra attempt of any question or any part of the question will not be considered.		

PART – II

- Q. No. 2.** Discuss evolution of vascular bundles in tracheophytes. (20)
- Q. No. 3.** Describe process of fertilization for an angiospermic plant with significance of double fertilization. (20)
- Q. No. 4.** Write a comprehensive account of Plant Growth Hormones. (20)
- Q. No. 5.** Give an account of structure of epidermis and describe various functions of its different cell types. (20)
- Q. No. 6.** Write a comprehensive account of light and dark reactions in plants. (20)
- Q. No. 7.** Describe various structural modification of roots, stems and leaves tissues in xerophytic plants. (20)
- Q. No. 8.** Write short notes on any FOUR of the followings:- (5 each) (20)
- (a) Theory of Natural Selection
 - (b) Economic importance of Rust and Smut
 - (c) Causes and significance of Polyploidy
 - (d) Types of stele
 - (e) Evolution of sporophyte in Bryophytes
